



PROJECT MAIEUTIC / 學問

From Inquiry to Software

The CityUHK Initiative on
Specification-Driven & Agentic Computing Education

“The art of bringing
latent knowledge
into light through
questioning.”



FROM FIRST YEAR ONWARD,
STUDENTS LEARN TO
**THINK, SPECIFY, AND
BUILD WITH AI AGENTS.**

TRACEBACK COACH

Helping Students Understand Python Errors Without Giving Away Solutions

Traceback Coach is an AI error-literacy coach for Jupyter notebooks. When a cell raises an error, it turns the traceback into a teaching moment—asking guiding questions, drawing a causal diagram, and giving just-enough hints. **It never shows the fix.**

★ **FIRST-YEAR
PROJECT**

Built by a Year-1
Computer Science Student

- Domain: Programming Education
- Tech Stack: Python, Jupyter, LLM API, TypeScript, Diagramming
- Mode: AI Coaching Agent
- Built With: Open Source

01 THE PROBLEM

```
Traceback (most recent call last):
  Cell In[7], line 4
    total += row["price"]
File ~/lib/report.py, line 118,
  in summarize(total, rows)
File ~/lib/core.py, line 402
  ...

NameError: name 'total'
is not defined
```

Forty lines of red, no explanation. So it gets pasted into a chatbot—and the lesson is lost.

THE COACH ASKS INSTEAD

“Where should **total** receive
its first value, before this
line?”

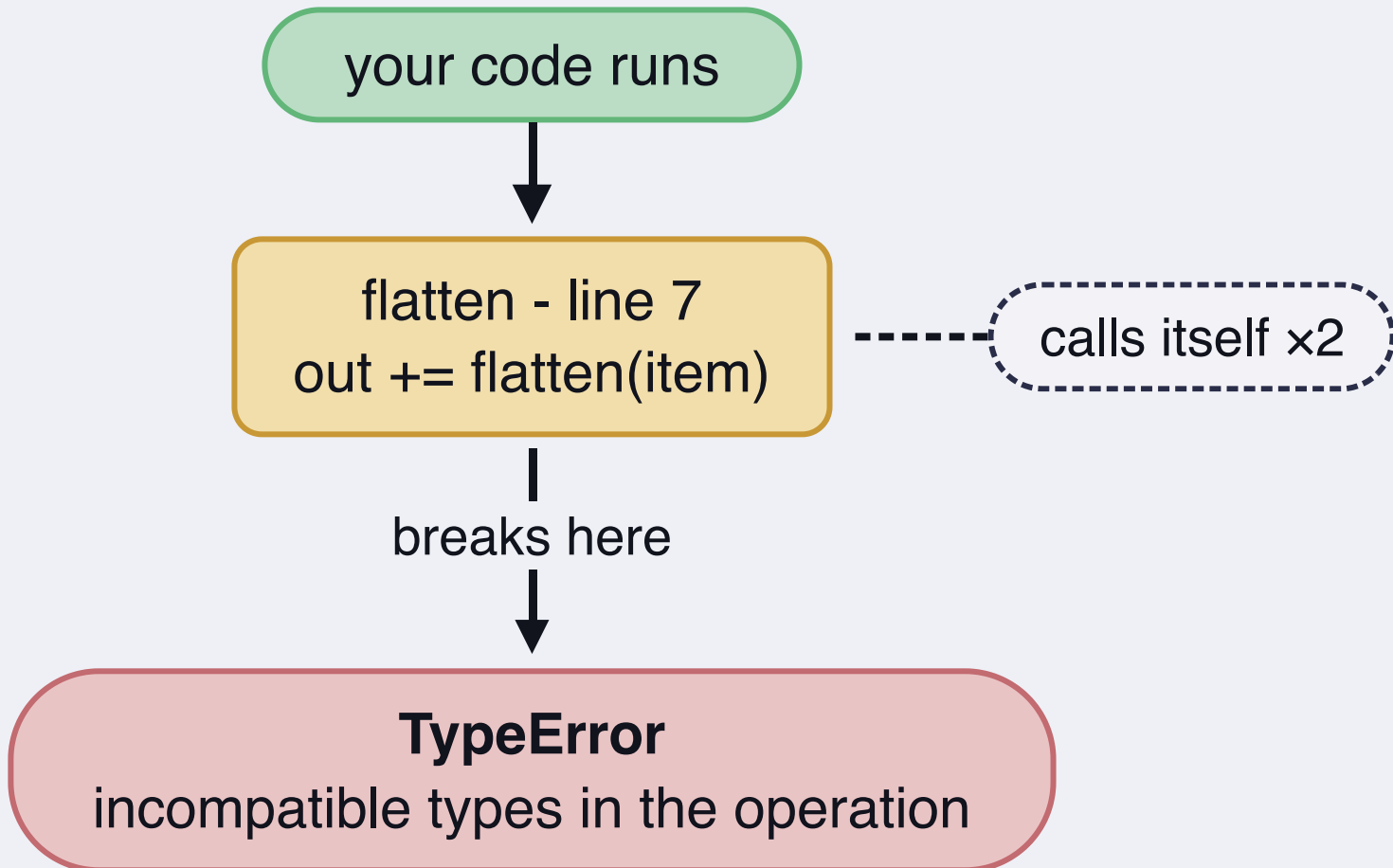
02 THE ANATOMY CARD

Fig. 1 – as a student sees it

Coach

What happened: You combined values whose types don’t go together: can only concatenate str (not "int") to str

Why it breaks:



- ★ **Where:** line 9 • out.append(item + 1) • falls when item is a str
- ★ **TypeError:** an operation got a type it can’t work with (e.g. str + int).
- ★ **Read it yourself:** The message names the clash — read ‘unsupported operand’ or ‘expected ...’
- ★ Use this error on a small example

? **Question:** What are the types of the two values on the failing line?

- one plain sentence, no jargon
- recursion drawn as a self-loop
- the exact failing line and token
- one guiding question, never a fix

03 HOW IT WORKS

- 1 Cell raises**
type • value • frames
- 2 Parse & clean**
noise frames dropped
- 3 Build the card**
diagram • example • question
- 4 Render inline**
never a fix

We teach students to
read errors. The AI
asks. They learn.

04 WHAT’S INSIDE

12 error families covered

0 network calls — fully offline

112 tests, headless core

EN + 繁體中文 the whole card, bilingual

- NameError
- TypeError
- ValueError
- IndexError
- KeyError
- AttributeError
- IndentationError
- SyntaxError
- ZeroDivisionError
- ImportError
- RecursionError
- UnboundLocalError
- ...

Fades as you learn
full → brief → nudge

Guess first
active-recall quiz

05 SESSION INSIGHTS

session_insights

You kept meeting the 'total' NameError — it’s common because the variable is only created when a condition runs. My hints nudged you to check the path before the loop and to initialize it early. Great job reading the message instead of copying code — that’s how real developers think.

Next step: try the quiz on NameError and another on TypeError.

Fig. 2 – session stats

Stats name the error family a student keeps losing to — then turn it into a question to sit with, not a fix to copy.

```
%coach
%coach_watch_on
%coach_lesson_IndexError
%coach_lang_zh
```

“ An expert is a person who has made all the mistakes that can be made in a very narrow field.

– NIELS BOHR

REFERENCES

Zhao, C., Wang, Z., Ahmedani, A. A., **Karimtayev, B.**, Marcello-Raphael-Christiano, Lua, N. M. B., Khanza, R. & Chan, C. (2026). DIVE into agentic coding: Can you teach AI to teach you? In Proceedings of the IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE). Accepted. traceback-coach: Fig. 3(a).



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Demo & Code

github.com/VoixKz/traceback-coach



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Changing Education. Empowering Minds. Building the Future.

